VISHWARAJ PRAVIN KOLGE

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SUMMARY

Mechanical Engineer with 5+ years of experience in the full product lifecycle development of automotive and heavy machinery products. Leveraged CAD, FEA, CFD, and PLM tools to deliver engineering project milestones through APQP process. Delivered 50% cash-flow retention and $180K in annual cost savings through value engineering initiatives. Interfaced with customers, cross-functional global teams, and suppliers, translating requirements into final product designs.

EXPERIENCE AND ACHIEVEMENTS

White Drive Motors and Steering LLC Hopkinsville, KY, USA
Design Engineer Apr 2024 – Present

- Accelerated new business pipeline by partnering with Sales and Applications team to develop custom motor variants
- Developed new motor load curves using Python and performed fluid pressure drop CFD analysis using ANSYS Fluent
- Retained 50% cash-flow during supplier disruptions by managing a sub-assembly insourcing initiative including design changes, project timeline, CAD models, BOM restructuring, team coordination, and PPAP validations
- Resolved critical customer claims with cross-functional 8D problem solving team, isolating root-cause to valving error through functional testing and implementing systemic fixes across affected process and inspection methods
- Achieved a 30% reduction in COGS by streamlining risk-based engineering change management, leveraging Siemens NX, Teamcenter PLM, ANSYS, GD&T, ProjectLibre, and SAP
- Reduced production disruptions and interpretation ambiguity by updating drawings to ASME Y14.5 GD&T standards

T.RAD North America Inc. Hopkinsville, KY, USA
Design Engineer Apr 2021 – Apr 2024

- Delivered 6 automotive heat exchanger design packages for OEMs, adhering to APQP phase-gate process while preparing 3D models and 2D drawings in CATIA/Creo with BOM management in IFS ERP
- Identified and mitigated risks for potential failure modes by executing 6 comprehensive DFMEA cycles
- Minimized verification cycle times by performing FEA and CFD simulations in ANSYS to assess product performance against thermal, vibrational, and pressure loads
- Spearheaded prototype verifications, including test plan development and execution, identifying and resolving technical failures and implementing corrective actions into mass production ready designs
- Secured 98% first-pass approval on 56 engineering changes, minimizing rework and preventing implementation delays
- Realized $180K in annual savings by spearheading VA/VE initiatives through strategic design and material optimization
- Resolved DFM/DFA constraints and reduced assembly cycle times by integrating design poka-yokes for error-proofing and replacing screw-fastened plastic tanks with snap-fit designs
- Orchestrated global technical alignment through weekly customer interactions and cross-functional collaboration across USA, Japan and Indian teams, ensuring integration of evolving product requirements into design cycles
- Enhanced team technical proficiency by developing 3 CAE Work Instructions and delivering 5 product training sessions
- Aided in retaining IATF 16949 certification by showcasing implementation of product development process to auditors

University of North Carolina at Charlotte – Computational Mechanics Lab Charlotte, NC, USA
Researcher Jul 2020 – Apr 2021

- Conducted comprehensive research on crack propagation prediction in composites using Finite Element Methods
- Generated 100+ FEA models in ABAQUS using XFEM and cohesive elements
- Automated modeling processes with Python scripting to generate unique model variations per simulation run

SKILLS

- **Engineering:** Thermal Management, Heat Transfer, Hydraulics, Strength of Materials, Machine Design, FEA, CFD
- **Technical:** APQP process, computational product verification, simulation-driven design and optimization, specification and test plan development, prototype testing and verification, failure resolution, VA/VE, CAD modeling, drawing preparation, BOM management, ASME Y14.5 GD&T, design package preparation, technical presentation
- **Tools:** NX, SolidWorks, Creo, CATIA, SAP, ANSYS, Python, MATLAB, Minitab, Teamcenter, ProjectLibre, Office
- **Manufacturing:** Metal forming and stamping, plastic injection molding, casting, machining, surface and heat treatment
- **Problem-Solving:** Failure Mode and Effects Analysis (FMEA), 8 Disciplines (8D), Root Cause Analysis (RCA)
- **Project Management:** Task scheduling and tracking, timeline management, Product Lifecycle Management (PLM)

EDUCATION

University of North Carolina at Charlotte Charlotte, NC, USA
Master of Science (MS) in Mechanical Engineering – GPA: 4.0 / 4.0 Aug 2018 – May 2020

Internship: Enventys Partners LLC Charlotte, NC, USA
Product Development Intern May 2019 – Aug 2019

- Designed, built, and tested prototypes for closet lighting IoT device, contributing to successful launch at CES 2020
- Developed electronic enclosure designs using SolidWorks, employing rapid prototyping techniques

Low Cycle Fatigue (LCF) evaluation tool, Industrial Design Project – Quest Global

- Developed a functional strain-based fatigue life evaluation tool using Python and resolved critical system errors
- Enhanced fatigue prediction accuracy by implementing Neuber and mean stress corrections and the Newton-Raphson iterative method into calculation scripts

Heat Conduction FEA of Chimney, Course – Finite Element Analysis

- Developed heat conduction FEM model in MATLAB by implementing Gauss Quadrature integration technique, achieving < 1% error when validated against ANSYS results

Statistical experimental design for burnt calorie estimation methods, Course – Design of Experiments

- Executed controlled full factorial experimental design to compare calorie-burnt metrics between gym bicycle sensors and smartwatch data; incorporated Minitab to randomize data runs and analyze collected data

University of Mumbai Mumbai, MH, India
Bachelor of Engineering (BE) in Mechanical Engineering – CGPA: 8.0 / 10.0 Jul 2013 – May 2017

Design and Fabrication of Go-Karts, Competitions – GKDC, IKR

- Achieved 2nd and 3rd place for 2 consecutive years in national competitions, showcasing exceptional teamwork
- Optimized critical chassis, sprocket, and steering components for weight and cost using SolidWorks design skills
- Facilitated vehicle assembly and testing through hands-on machine shop experience aiding operational readiness

CERTIFICATIONS

- **Advanced GD&T ASME Y14.5 v2009,** Applied Geometrics Inc. Nov 2025
- **Fundamentals of GD&T ASME Y14.5 v2018,** Applied Geometrics Inc. Jul 2024
- **Six Sigma: Define and Measure,** edX Feb 2021
- **GD&T and Stack-Up,** Udemy Nov 2020
- **Risk Management: Master FMEA/FMECA & Criticality from A to Z,** Udemy Oct 2020
- **A Hands-on Introduction to Engineering Simulations,** edX Jul 2020